Eureka Technology

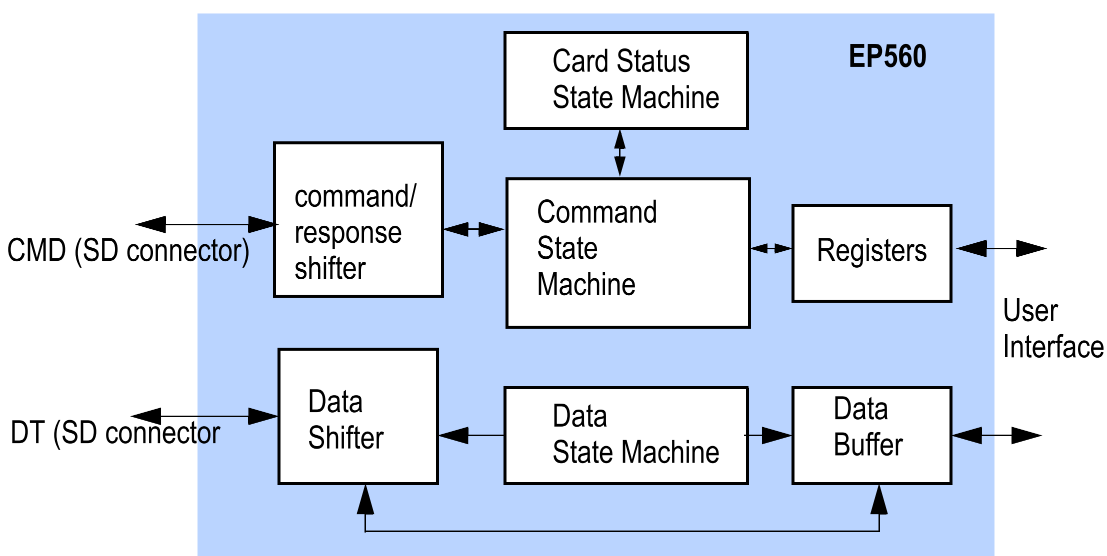
EP560 SDIO/SD memory Slave Controller

Data Sheet

FEATURES

- Slave controller for SDIO, SD memory card.
- Compatible with SD specification 2.0 and SDIO 1.2.
- Provides SD interface to peripheral or memory device through a simple address/data interface.
- Support SD and SPI bus protocol.
- Option to integrate with other standard CPU buses such as AMBA AHB, PowerPC, MIPS, etc.
- Maximum transfer rate of 25Mbyte/sec at SD interface.
- Simple 32-bit user interface.
- Write buffer can post up to 4086 bytes for high performance block transfer.
- Supports maximum block size up to 2048 bytes.
- Process most SD/SDIO commands automatically without user interference.
- Contains SD memory/SDRIO standard slave register set.
- Hardware CRC generation and detection.
- Supports multi-function SD cards, command suspend, resume, block transfers, and SDIO interrupts.
- Static inputs allow user to define application specific register data.
- Designed for ASIC and FPGA implementations.
- Fully static design with edge triggered flip-flops.

BLOCK DIAGRAM



DESCRIPTION

The SD Slave Controller is designed to reside within an SD memory or SDIO card. It serves as an interface between the SD bus and user logic that provides the actual function of the card. It is designed to integrate with user logic to make various devices using the SD bus protocol, such as storage or wireless network card.

The SD slave controller supports both 1 and 4 bit SD interface mode and SPI mode. Data rate of up to 25Mbyte/sec (200Mbs) can be realized with SD interface. Features such as plug and play, auto-detection, error correction, write protection are standard with SD card interface and are supported.

As a slave device, the SD slave controller received commands from the host through the SD interface. Most of the commands are processed locally by the controller without any help from the user logic. The standard register set is also implemented within the slave controller.

In case of memory or IO access that needed to be forwarded to the user logic, the slave controller handles all the SD bus protocol and presents the request to the user logic as simple read and write request through parallel address and data buses. Burst transfer of up to 2048 bytes per transfer and user defined wait states are supported on the user interface to maximize data bandwidth. The slave controller also contains data buffer to match the speed differences between the user interface and the SD interface. It allows a much more efficient use of the user interface.

With the EP560, SD card design can be realized with very little development cost. Designer can add SD memory and SDIO interface capability to the design by simply adding the EP560 module without changing the rest of the system architecture.

1 Signal Descriptions

The following tables listed all the external I/O signals of SD slave controller. All signals with “#” or “_B” suffix are active low signals.

The following table describes the SD card interface signals between the controller and the SD connector

Table 1: SD Card Interface

Name	Type	Descriptions
SDCLK	I	Clock input from SD card interface.
CMD	I/O	Command and response on SD interface.
DT[3:0]	I/O	Data line. The EP560 supports 1 bit and 4 bit data transfer

The following table describes the generic user interface signal for the user to access the SD controller.

Table 2: User Interface Signals

Signal	Type	Description
ADDR[31:0]	O	Memory or IO address to the user logic
ADS#	O	Address strobe. Output to the user interface to start an access. The access type are provided in the TYP and WR signals. Other information such as address, byte enable and size information are valid from ADS# assertion to the last data transfer acknowledge.
BE#[3:0]	O	Byte enable. Indicates the byte(s) to be transferred.
BLAST#	O	Burst Last. This signal is asserted by the controller to the user interface to indicate the last data transfer of a burst sequence. For single data access, BLAST# is asserted as the same time as ADS#.

Table 2: User Interface Signals

Signal	Type	Description
DRD[31:0]	I	Read data. This bus provides the data read from the user interface to the controller core.
DWR[31:0]	O	Write data from the controller core to the user interface.
ERA_PARAM_ERR	I	Status input. Indicates invalid selection or write blocks for erase operation.
FUNC_NUM[2:0]	O	SDIO function number of the current access
RDY#	I	Data ready or command acknowledge. This is the response signal from the user interface in response to ADS# assertion. For commands that involve data transfer, this signal indicates a valid data transfer on the user interface. For read access, valid read data is presented on DRD at the same time as RDY#. For write access, DWR is accepted by the core when RDY# is asserted. For burst data transfer, RDY# is asserted for one cycle for each data word. For commands that does not involve data transfer, the assertion of RDY# indicates that the command is received by the user logic.
SIZE[9:0]	O	Transfer size. This signal indicates the number of 32-bit to be transferred for the ADS# command.

Table 2: User Interface Signals

Signal	Type	Description
TYP[2:0]	O	Command type. This signal is used together with ADS# to indicate the command send to the user logic. 000 - Normal data read or write to memory as indicated by WR. 001 - Normal data read or write to IO as indicated by WR. 010 - SD status register access. This indicates reading from the SD status register instead of the normal data read. 011 - CSA read or write to indicated by WR. 100 - Erase start address. Indicate to user logic to mark the value in ADDR output as the erase start address. 101 - Erase end address. Indicates to user logic to mark the value in ADDR output as the erase end address. 110 - Erase execution. Indicates to user logic to start erase operation. 111 - Reserved. Only command types 000 to 011 has data transfer. Other command types does not involve data transfer.
WP_ERA_SKIP	I	Status input. Indicates partial address was erased due to write protect.
WP_VIOL_ERR	I	Status input. Indicates attempt to write to write protected block.
WR	O	Output to user interface to indicate read or write access. This signal is high for write and low for read.

Table 3: System and Static Signals

Signal	Type	Description
CCCR[144:0]	I	Static input. Lower 145 bits of Card Common Control Registers read only values for SDIO
CCCR_OUT [145:17]	O	Output from CCCR register, reflect the values of the corresponding bits in the CCCR register. Only the following bits which are writable by the host have valid information. All other bits are hard-wired to “0”. The valid bits are: 145, 143:128, 107:104, 97, 69, 63, 61, 57:56, 51:48, 39:32, 23:17.
CCCR_UPP ER[127:0]	I	Static input. Upper most 128 bits of Card Common Control Registers read only values for SDIO
CID[127:0]	I	Static input. Card Identification Register (CID) read only values, for SD memory.
CSD10[127: 0]	I	Static input. Card Specific Data v1.0 Register read only values, for SD Memory.
CSD20[127: 0]	I	Static input. Card Specific Data v2.0 Register read only values, for SD Memory.
DSR[15:0]	O	Output of the Driver Stage Register. DSR is written by the host as SD memory command.
EPS	O	Output from the FBR register reflecting the value of the EPS bit (bit 17)
FBR1[143:0]	I	Static input. Function Basic Register read only values for SDIO Function 1.
HRESET#	I	Hardware reset. Indicates when power is stable and the controller can start operation.
INTR_LINE	I	Interrupt request from user logic. This signal is level sensitive. Once asserted, it should remain asserted until the interrupt condition no longer exist.

Table 3: System and Static Signals

Signal	Type	Description
OCR_SDIO[23:0]	I	Static input. Operation Condition Register read only values, for SDIO.
OCR_SDM[31:0]	I	Static input. Operation Conditions Register read only values, for SD memory.
SCR[63:0]	I	SD COnfiguration Register read only values, for SD memory.
SDIO_CRD_ RDY	I	SDIO card initialization. This signal is high to indication initialization complete and the card is ready.
SRESETI#	O	SDIO soft reset output
SRESETM#	O	SD memory soft reset output

2 User Interface

This section describes the operation of the user interface. The user interface operates at the domain of the SD Clock (SDCLK). All inputs are sampled with the rising edge of SDCLK and all output signals are driven from the rising edge of SDCLK.

Most of the interaction between the SD bus and the controller is handled by the SD slave controller automatically without any action required by the user logic. Only a few transactions such as memory read, write or erase are forwarded to the user logic through the user interface.

The following sections describes all of those interactions between the SD slave controller and the user logic. SD bus protocol or commands not described in this section are handled internally by the slave controller thus not a concern of the user interface.

2.1 User Commands and Basic Protocol

SD Slave and User interaction is initiated by the SD Slave asserting ADS# with a valid command. The command is on the TYP signal. The commands can be a data transfer request or an erase command. The RDY# is asserted by the user logic when data is ready for data transfer or when the command has been acknowledged and that the response is valid. The RDY# assertion timing depends on the type of command. For timing and details refer to the sections below.

2.2 Single Read

All read accesses are originated from the SD interface by the system host. The slave controller receives the command from the host and generate the proper responses to the host and performs CRC checking automatically. If it is a read access to user area, the controller forwards the read request to the user logic on a block by block basis. Read data returned from the user logic are first stored in the controller and then transferred to the SD interface with CRC generated automatically by the controller. All SD bus protocol, including start bit, end bit, etc., are handled by the controller.

A read operation to the user interface is initiated by the SD slave by asserting ADS#. TYP signal is "000" to indicate memory access, “001” to indicate IO access and “011” to indicate CSA access. The WR signal is low to indicate a read access. BLAST# is asserted to indicate single read. The address to access is on the ADDR signal. SIZE is 001 (hex) to indicate one word is being accessed. If it is an SDIO operation (TYP = “001”), the FUNC_NUM signal indicates which function the command is accessing. All the fore mentioned signals are valid at the same time that ADS# is asserted and remain valid until the end of the transfer.

After ADS# is asserted, user may proceed with the read operation and returns the read data to the slave controller when it is ready. RDY# is asserted by the user logic to signal read data is available on the DRD signal. The number of clock cycles between ADS# assertion and RDY# assertion is controlled by the user but must be within the time limit specified below.

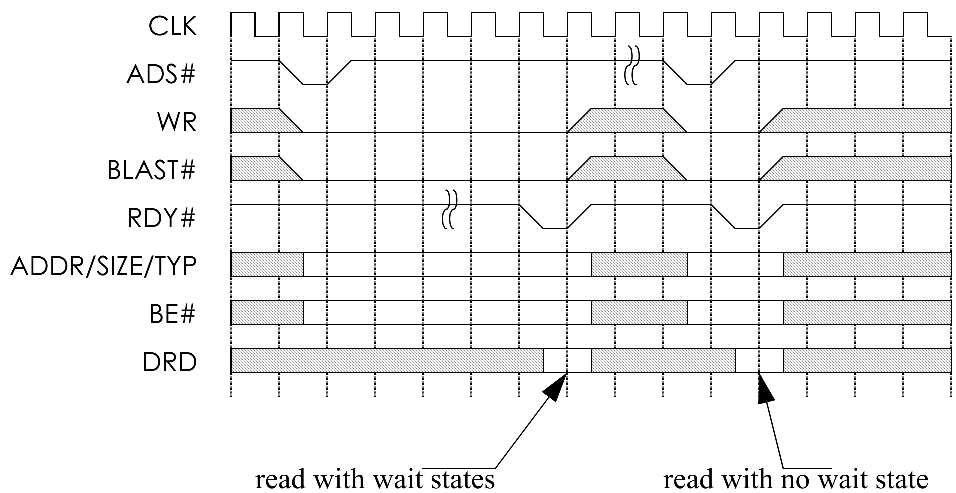


Figure 2.1 Single data read access

2.3 Burst Read

A burst read access is very similar to that of a single read. The read access is initiated in the same exact way as the single read access, except that SIZE is not 001 (hex) but the number of blocks to be transferred and BLAST# is asserted at the last data phase of the burst read. RDY# is asserted by the user logic when each read data is available on the DRD signal. User may assert RDY# continuously or insert wait state in between subsequent data if needed by de-asserting RDY# in between data transfers. The transfer ends when the number of data being transferred equals the number specified in SIZE. The controller also asserted BLAST#

to signal the last data phase.

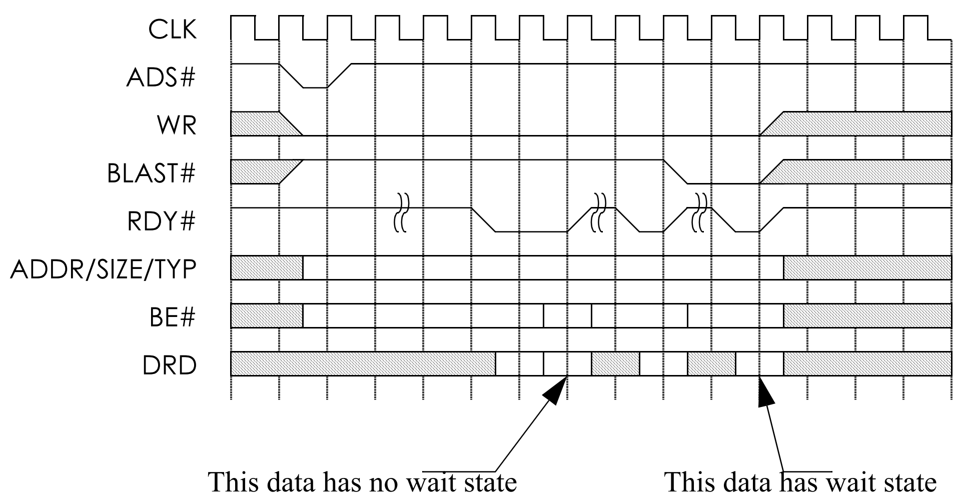


Figure 2.2 Burst data read access

2.4 Read Access Time Limit

For single or burst data read access to standard capacity SD Memory and SDIO cards, RDY# of the last data must be asserted within a certain time limit from ADS# assertion. The time limit is

((100 * ((TAAC * fpp) + (100* NSAC))) - 1) cycles or 100 ms, whichever is less.

The TAAC and NSAC values are defined in the CSD register in later section. The value of TAAC can range from 1ns to 80ms. NSAC can range from 0 to 255. fpp is the frequency of the SD clock.

For the read access of high capacity SD Memory cards RDY# of the last data must be asserted within 100 ms. from ADS# assertion.

2.5 Single Write

All write accesses are originated from the SD interface by the system host. The slave controller receives the command from the host and generate the proper responses to the host and performs CRC checking automatically. If it is a write access to user area, the controller first post each block of write data into its write buffer, checks CRC and then forwards the write request to the user logic on a block by block basis. The controller can buffer up to 4096 bytes of data and capable of writing data to the user logic at the same time receiving data from the host. All SD bus protocol, including status bits, busy, etc., are handled by the controller.

A single write operation is similar to the single read operation. The involved signals are the same at the time ADS# is asserted by the SD Slave, with the exception of WR, which is high to indicate a write operation, and data is valid on DWR signal. RDY# is asserted by the user logic when is accept for write. Similar to single read, the transfer ends when RDY# is asserted. The number of clock cycle between ADS# assertion and RDY# assertion is controlled by the user.

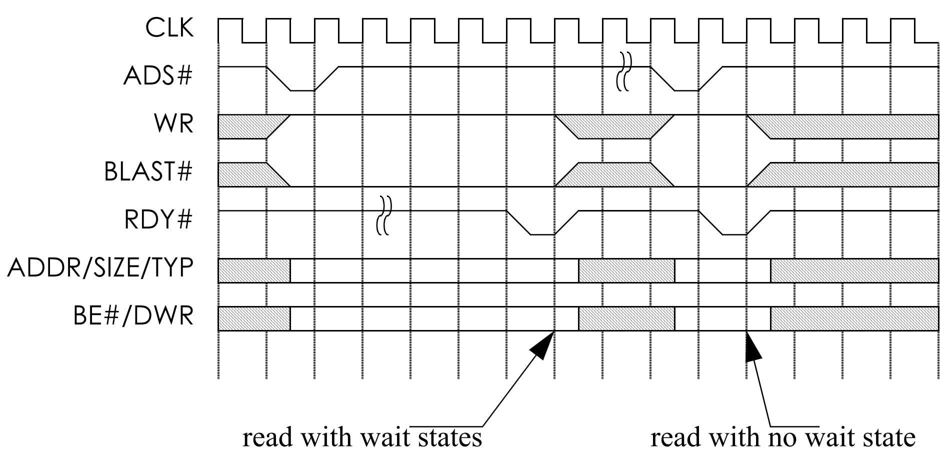


Figure 2.3 Single data write access

2.6 Burst Write

A burst write operation is similar to burst read as single write is to single read. The involved signals sent at the time ADS# is asserted by the SD Slave as in the case of single write, except BLAST# and SIZE. SIZE indicates the number of 32-bit words to be transferred and BLAST# will be asserted when the last 32-bit word is sent on DWR. SIZE is valid from the time ADS# is asserted until the end of the transfer. RDY# is asserted by the user logic when the data is accepted. The controller drives the next write data on the DWR in the cycle following RDY# assertion. RDY# may be asserted continuously by the user if no wait state is needed or insert wait states in between RDY# assertion. BLAST# is asserted by the SD Slave when the last word is sent on DWR and the transfer ends when both BLAST# and RDY# are asserted at the same cycle.

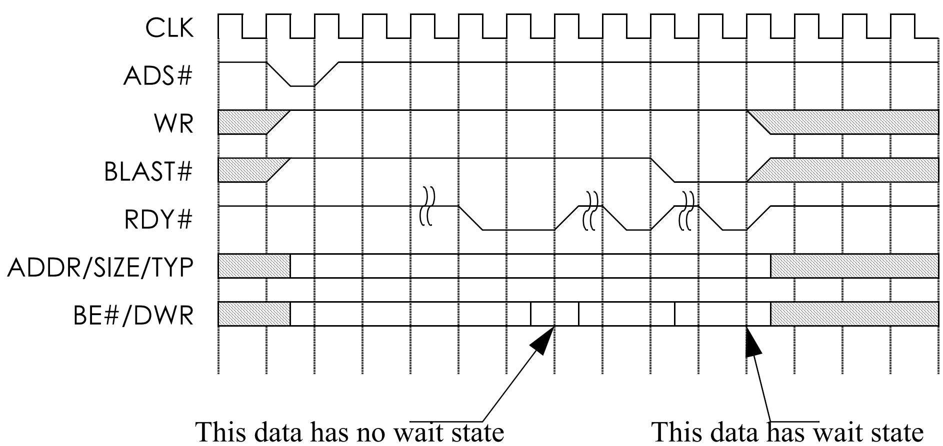


Figure 2.4 Burst data write access

2.7 Write Access Time Limit

For write access of a standard capacity SD Memory and SDIO cards, RDY# of the last data must be asserted within a certain time limit from ADS# assertion. The time limit is the lessor of the following two limits:

1. the read access time limit multiplied by R2W_FACTOR, or

2. 250ms

The read access time limit is defined in previous section. R2W_FACTOR is defined in CSD register and has a range from 1 to 32.

For high capacity SD Memory cards, RDY# shall be asserted within 250 ms.

2.8 Erase

An erase operation is executed through a sequence of three commands by the SD Slave: erase write block start, erase write block end, and erase execute. To start the sequence, the ADS# is asserted by the SD Slave. At this time the TYP signal has the value "100" and on the ADDR signal will be the address of the first write block to be erased. The user logic must verify that the address provided is valid. When RDY# is asserted by the user logic, ERA_PARAM_ERR is asserted by the user if the address is invalid. ERA_PARAM_ERR is de-asserted if the address is valid. The RDY# signal must be asserted within 62 clock cycles from the time ADS# is asserted.

The second command in the sequence, erase write block end, is handled very similarly to the first. The ADS# is asserted by the SD Slave, but the value on the TYP signal is "101". The value on ADDR signal indicates the last write block of the continuous range to be erased. The user logic must verify that the address provided is valid. When RDY# is asserted by the user logic, ERA_PARAM_ERR is asserted by the user if the address is invalid. ERA_PARAM_ERR is de-asserted if the address is valid. The RDY# signal must be asserted within 62 clock cycles from the time ADS# is asserted.

The final command is erase execution. The ADS# is asserted by the SD Slave and the value on the TYP signal is "110". At this time the user logic begins erasing the blocks in the address range as provided on the previous erase start and erase end commands. The RDY# signal is asserted by the user logic when erase has completed. If blocks were skipped during erase due to the fact they were write protected, WP_ERA_SKIP is asserted when RDY# is asserted by the user logic. If no blocks were skipped due in erase execution, WP_ERA_SKIP is de-asserted.

The SD bus specification requires that the RDY# for erase execution must be asserted within 250ms from ADS# assertion.

The 3 command erase sequence is issued by the host to the slave controller. Receiving the erase address start or erase address end commands does not guarantee the erase execute command will be issued. If any of the first two commands results in error, the host may re-start the sequence again without finishing the previous sequence. User logic should erase its memory only at the receipt of the erase execute command.

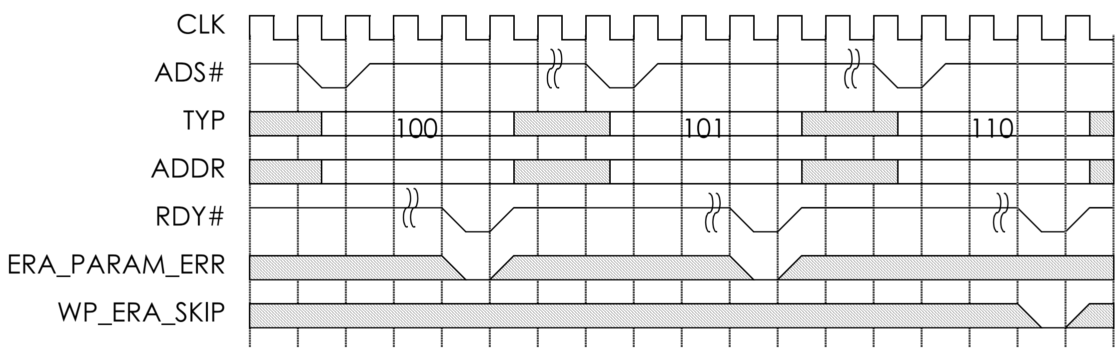


Figure 2.5 Erase Sequence

2.9 Reset

The controller core should be reset by the RESET# pin during initial power up. In addition, soft reset can be executed by the host by sending a command to the SD Slave. The soft reset is forwarded by the SD Slave to the user logic by asserting either the SRESETI# or SRESETM# signal. Soft reset commands are card type specific (i.e. SD Memory or SDIO). SRESETI# indicates SDIO reset and SRESETM# indicates SD memory reset.

2.10 Initialization

During initial power up of SDIO device, the user logic may require a long time to reach operating stage. The SDIO_CRD_RDY signal can be used by the user logic to indicate its readiness. It is asserted by the user logic so that the proper response can be sent to the host by slave controller to indicate the status of the user logic. For SD Memory device, this is done through bit 31 of the OCR input to the controller core.

2.11 Interrupts

Interrupts are available for SDIO and may be sent at anytime by the user by asserting INTR_LINE. Once it is asserted it should remain asserted until the interrupt condition is handled.

Upon INTR_LINE assertion, the SD Slave forwards the request to the host at the interrupt window allowed by the SD bus protocol. In 4-bit transfer mode, the request can only be send at specific interrupt window when the data bus is not used. In 1-bit transfer mode, interrupt is sent to the host immediately.

2.12 Register Read

All the SD registers are contained internally in the slave controller except the SD Status Register (SSR). The SD status register contains 512 bits of mostly read only data so it is more appropriate to be stored in the user logic.

When the host issues a command to read the SD Status Register, the controller forwards the read request to the user logic through a regular burst read transfer. except that the TYP signals is "010" to indicate SD Status Register read.

The SIZE signal is always 010(hex) for SSR read to indicate 512 bits (16 words) transfer. ADDR signal is always 00000000(hex) to indicate starting reading from word 0 (bit 31 to 0) of the SSR. WR is also "0" to indicate read access.

Bit 511 and 510 of the SSR are generated internally by the SD slave controller because it contains information known to the slave controller. When reading the last word of the SSR, the user can provide either "1" or "0" in this two bits and both will be substituted by the slave controller when forward to the host.

2.13 Static Inputs

Except for SD Status Register, all the other registers are contained internally in the slave controller. A number of read-only register data such as Manufacturer ID is provided to the SD Slave via static inputs which can be hardwired by the user to the proper value. Some registers are completely self-contained within the controller core and do not require static inputs. For each register requiring static value from the user, there is a dedicated static input signal. The bit index to the static input matches the bit index of the corresponding bit in the register.

Most registers contain a mixture of user provided signals, reserved bits and contents generated from within the SD Slave controller. In this case, the reserved bits and SD slave controller generated bits will be forwarded to the SD host. The corresponding bit slice on the static input shall be tied to either "1" or "0" and will be ignored by the controller. For more details about the SD Registers, see section 4.

3 SD Registers

The SD controller contains most of the required SD registers. Access to these registers is processed by the SD slave controller automatically and the access is not forwarded to the user interface. The SD Status Register (SSR) is the only exception. Because this register contains 512 read only bits, it is more efficient that these read only data bits are stored in the user logic and provided to the slave controller through the register read command described in previous section.

The following table lists all the SD registers. The CSR, DSR and RCA is completely internal to the controller core and is not dependent on as static inputs from the user. Other register except for SD Status use some of the value from the static inputs provided by the user.

Table 4: SD Registers

Register	Source
OCR	Internal and static input
CID	Internal and static input
CSD version 1.0	Internal and static input
CSD version 2.0	Internal and static input
SCR	Internal and static input
SD Status	User Logic
SDIO OCR	Internal and static input
CCCR	Internal and static input
FBR	Internal and static input
CSR	Internal to controller core
DSR	Internal to controller core
RCA	Internal to controller core

3.1 OCR Register

The OCR consists of 32-bits of read only values. The read only value are static input to the core through the OCR_SDM inputs. The following table lists the bits in this register.

Table 5: OCR Registers

Bits	Description	Source
31	Card power up status bit	Input from OCR_SDM
30	Card capacity status	Input from OCR_SDM
29:24	Reserved	Generated by the controller
23:15	VDD voltage	Input from OCR_SDM
14:8	Reserved	Generated by the controller
7	Low voltage range	Input from OCR_SDM
6:0	Reserved	Generated by the controller

3.2 CID Register

The CID consists of 128-bits of read only values. The read only value are static input to the core through the CID inputs. The following table lists the bits in this register.

Table 6: CID Registers

Bits	Description	Source
127:120	Manufacturer ID	From CID input signal
119:104	OEM/Application ID	From CID input signal
103:64	Product name	From CID input signal
63:56	Product revision	From CID input signal
55:24	Product serial number	From CID input signal
23:20	Reserved	Generated by the controller
19:8	Manufacturing date	From CID input signal
7:1	CRC7 checksum	From CID input signal
0	Reserved	Generated by the controller

3.3 CSD Version 1.0 Register

This register has 128 bits and is needed for version 1.0 SD card. The read only value are static input to the core through the CSD10 inputs. The following table lists the bits in this register.

Table 7: CSD Version 1.0 Registers

Bits	Description	Source
127:126	CSD structure	From CSD10 input signal
125:120	Reserved	Generated by the controller
119:112	TAAC	From CSD10 input signal
111:104	NSAC	From CSD10 input signal
103:96	TRAN_SPEED	From CSD10 input signal
95:84	CCC	From CSD10 input signal
83:80	READ_BL_LEN	From CSD10 input signal
79	READ_BL_PARTIAL	From CSD10 input signal
78	WRITE_BLK_MISALIGN	From CSD10 input signal
77	READ_BLK_MISALIGN	From CSD10 input signal
76	DSR_IMP	From CSD10 input signal
75:74	Reserved	Generated by the controller
73:62	C_SIZE	From CSD10 input signal
61:59	VDD_R_CURR_MIN	From CSD10 input signal
58:56	VDD_R_CURR_MAX	From CSD10 input signal
55:53	VDD_W_CURR_MIN	From CSD10 input signal
52:50	VDD_W_CURR_MAX	From CSD10 input signal
49:47	C_SIZE_MULT	From CSD10 input signal
46	ERASE_BLK_EN	From CSD10 input signal
45:39	SECTOR_SIZE	From CSD10 input signal

Table 7: CSD Version 1.0 Registers

Bits	Description	Source
38:32	WP_GRP_SIZE	From CSD10 input signal
31	WP_GRP_ENABLE	From CSD10 input signal
30:29	Reserved	Generated by the controller
28:26	R2W_FACTOR	From CSD10 input signal
25:22	WRITE_BL_LEN	From CSD10 input signal
21	WRITE_BL_PARTIAL	From CSD10 input signal
20:16	Reserved	Generated by the controller
15	FILE_FORMAT_GRP	Generated by the controller
14	COPY	Generated by the controller
13	PERM_WRITE_PROTECT	Generated by the controller
12	TMP_WRITE_PROTECT	Generated by the controller
11:10	FILE_FORMAT	Generated by the controller
9:8	Reserved	Generated by the controller
7:1	CRC	Generated by the controller
0	Reserved	Generated by the controller

3.4 CSD Version 2.0 Register

This register has 128 bits and is needed for high capacity (version 2.0) SD card. The read only value are static input to the core through the CSD20 inputs. The following table lists the bits in this register.

Table 8: CSD Version 2.0 Registers

Bits	Description	Source
127:126	CSD structure	Generated by the controller
125:120	Reserved	Generated by the controller

Table 8: CSD Version 2.0 Registers

Bits	Description	Source
119:112	TAAC	Generated by the controller
111:104	NSAC	Generated by the controller
103:96	TRAN_SPEED	From CSD20 input signal
95:84	CCC	From CSD20 input signal
83:80	READ_BL_LEN	Generated by the controller
79	READ_BL_PARTIAL	Generated by the controller
78	WRITE_BLK_MISALIGN	Generated by the controller
77	READ_BLK_MISALIGN	Generated by the controller
76	DSR_IMP	From CSD20 input signal
75:70	Reserved	Generated by the controller
69:48	C_SIZE	From CSD20 input signal
47	Reserved	Generated by the controller
46	ERASE_BLK_EN	Generated by the controller
45:39	SECTOR_SIZE	Generated by the controller
38:32	WP_GRP_SIZE	Generated by the controller
31	WP_GRP_ENABLE	Generated by the controller
30:29	Reserved	Generated by the controller
28:26	R2W_FACTOR	Generated by the controller
25:22	WRITE_BL_LEN	Generated by the controller
21	WRITE_BL_PARTIAL	Generated by the controller
20:16	Reserved	Generated by the controller
15	FILE_FORMAT_GRP	Generated by the controller
14	COPY	Generated by the controller
13	PERM_WRITE_PROTECT	Generated by the controller

Table 8: CSD Version 2.0 Registers

Bits	Description	Source
12	TMP_WRITE_PROTECT	Generated by the controller
11:10	FILE_FORMAT	Generated by the controller
9:8	Reserved	Generated by the controller
7:1	CRC	Generated by the controller
0	Reserved	Generated by the controller

3.5 SCR Register

This register has 64 bits. The read only value are static input to the core through the SCR inputs. The following table lists the bits in this register.

Table 9: SCR Registers

Bits	Description	Source
63:60	SCR_STRUCTURE	From SCR input signal
59:56	SD_SPEC	From SCR input signal
55	DATA_STAT_AFTER_ERASE	From SCR input signal
54:52	SD_SECURITY	From SCR input signal
51:48	SD_BUS_WIDTHS	From SCR input signal
47:32	Reserved	Generated by the controller
31:0	Reserved for Manufacturer usage	From SCR input signal

3.6 SDIO OCR Register

This register has 24 bits. The read only value are static input to the core through the SDIO_OCR inputs. The following table lists the bits in this register.

Table 10: SCR Registers

Bits	Description	Source
23:8	VDD voltage	From SDIO_OCR input signal
7:0	Reserved	Generated by the controller

3.7 CCCR Register

This register has 256 bytes. The read only value are static input to the core through the CCCR and the CCCR_UPPER inputs. The following table lists the bits in this register.

Table 11: SCR Registers

Bits	Description	Source
2047:192 0	Manufacturer specific	From CCCR_UPPER input sig- nal
1919:146	Reserved	Generated by the controller
145	EMPC	Generated by the controller
144	SMPC	From CCCR input signal
143:128	IO block size for function 0	Generated by the controller
127:120	Ready flags	From CCCR input signal
119:112	Exec flags	From CCCR input signal
111	DF	From CCCR input signal
110:108	Reserved	Generated by the controller
107:104	Function select	Generated by the controller
103:98	Reserved	Generated by the controller
97	BR	Generated by the controller
96	BS	From CCCR input signal
95:72	CIS pointer	From CCCR input signal

Table 11: SCR Registers

Bits	Description	Source
71:64	Card capability	From CCCR input signal except bit 69 which is generated by the controller
63	CD disable	Generated by the controller
62	SCSI	From CCCR input signal
61	ECSI	Generated by the controller
60:58	Reserved	Generated by the controller
57:56	bus width	Generated by the controller
55:52	Reserved	Generated by the controller
51	RES	Generated by the controller
50:48	AS2:0	Generated by the controller
47:41	Interrupt pending	From CCCR input signal
40	Reserved	Generated by the controller
39:32	Interrupt enable	Generated by the controller
31:25	IO ready	From CCCR input signal
24	Reserved	Generated by the controller
23:17	IO enable	Generated by the controller
16:12	Reserved	Generated by the controller
11:8	SD specification revision	From CCCR input signal
7:0	CCCR/SDIO revision	From CCCR input signal

3.8 FBR Register

This register has 256 bytes. There is one FBR register for each function. The read only value are static input to the core through the FBR1 inputs. The following table lists the bits in this register.

Table 12: SCR Registers

Bits	Description	Source
2047:144	Reserved	Generated by the controller
143:128	IO block size for function 1	Generated by the controller
127:120	CSA window	Generated by the controller
119: 96	CSA pointer	Generated by the controller
95:72	CIS pointer	From FBR1 input signal
71:18	Reserved	Generated by the controller
17	EPS	Generated by the controller
16	SPS	From FBR1 input signal
15:8	Extended standard SDIO Function interface code	From FBR1 input signal
7	CSA enable	Generated by the controller
6	CSA support	From FBR1 input signal
5:4	Reserved	Generated by the controller
3:0	Standard SDIO Function interface code	From FBR1 input signal

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